



Distributed Databases: Homogeneous and Heterogeneous Databases

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Distributed Databases

- A distributed database is a database that is not stored in just one computer.
- Instead, it is spread across many computers or different locations.
- These computers are connected through a network (like the Internet or a private network).
- Even though the database is stored in many places, it should look like one single database to the users.
- Each location stores a part of the data or sometimes a copy of the whole data.
- The different locations do not share the same hardware (they are physically separate).

Why use distributed databases?

- → When many users from different parts of the world need to access the database.
- → To make the system faster, more reliable, and easily available even if one location fails.
- Management is important:
The database must be managed properly so users don't feel like they are connecting to different places.

ADVANTAGES

- Faster access to data (data closer to users).
- If one system fails, others can still work (fault tolerance).
- Easier to scale (can add more computers as needed).

DISADVANTAGES

- **Complex to manage:**

Managing many databases in different locations is much harder than managing a single one.

- **High cost:**

Setting up and maintaining a distributed system can be expensive (need good network, hardware, and software).

- **Data security risks:**

Data travels over networks, so it can be attacked if not protected properly.

- **Difficult to keep data updated:**

When data is changed in one location, it must be updated in all others — this can be tricky and slow.

Examples:

- Banking systems (branches in different cities sharing customer data).
- Cloud services (like Google Drive, Amazon databases).

summary

"Distributed databases are very useful but need careful management.

If not handled well, they can cause problems like high costs, security risks, and data mismatch."

Distributed databases are divided into 2 :

1. Homogeneous Distributed Database

2. Heterogeneous Distributed Database

1. Homogeneous Distributed Database

Advantages:

- All sites (computers/servers) in the system use the same type of database management system (DBMS).
 - The database structure (how data is organized) is also the same at all sites.
 - All sites are aware of each other and communicate easily.
 - They cooperate fully to handle user requests and transactions.
- Easy to coordinate.
 - Fast processing.
 - Easier to secure and update.

Example:

- A bank has branches in different cities, and all branches use Oracle DBMS with the same database design to manage customer data.

2. Heterogeneous Distributed Database

- Different sites use different types of DBMS software.
- The database structure may be different at different sites.
- Sites may not know about each other completely.
- Cooperation between sites is limited and more difficult.

Challenges:

- Need special software (middleware) to help them talk to each other.
- More chances of data mismatch or errors.
- Slower transaction processing.

Example:

- An international company where offices in different countries use different database systems because of local requirements.

Why Homogeneous is easier?

Because everything is the same, so less confusion!

Why Heterogeneous is flexible?

Because different sites can choose the best database for their needs.

But...

Heterogeneous systems need more effort to keep everything working together properly.

Conclusion :

"Homogeneous distributed databases are easy and fast because everything is the same.

Heterogeneous distributed databases offer flexibility but are harder to manage because everything is different."



Thank
You